

CHAPTER FOUR

DECISION-MAKING

Meaning:

- Decision-making is a rational choice or selection of one alternative from among a set of alternatives; i.e. it is the act of choosing one alternative from among a set of alternatives.
- Decision-making is the management function that consists of choosing one course of action from all the available alternatives.

Decision-making is part of every aspect of the manager's duties, which include planning, organizing, staffing, leading and controlling, i.e. decision-making is universal. In all managerial functions decision-making is involved. All managerial functions have to be decided. For example, managers can formulate planning objectives only after making decisions about the organization's basic mission. Even though in all managerial functions decision-making is involved, the critical decision-making is during planning because planning identifies the objectives of the organization; i.e. decision must be made to identify the objectives/missions of an organization. In the planning process, managers decide such matters as what goals or opportunities their organization will pursue, what resources will be used, who will perform each required task etc. The entire planning process involves managers in a continual series of decision-making situations.

Decision-making has three elements (parts)

1. When managers make decisions; they are choosing or selecting from among alternatives.
2. When managers make decisions, they have available alternatives. When there are no alternatives, there is no decision-making, rather it become mandatory.
3. When managers make decisions, they have purpose in mind. The purpose in mind is organizational objectives.

THE DECISION-MAKING PROCESS

Decisions are organizational responses to problems. Every decision is the outcome of a dynamic process that is influenced by multitude of forces. So decision-making has its own processes / series of steps. The process is a sequential process rather than a series of steps.

1. Identifying problems

A necessary condition for a decision to exist is a problem - the discrepancy between an actual and desired state; a gap between where one is and where one wants to be. If problems do not exist, there will be no need for decisions; i.e. problems are prerequisites for decisions. How critical a problem for the organization is measured by the gap between levels of performance specified in the organization's goals and objectives and the level of performance attained; i.e. it is measured by the gap between level of performance specified (standards set) and level of performance attained. The problem is very critical when the gap between the standard set and actual performance attained is very high. To locate problems, managers rely on several different indicators:

- Deviations from past performance. A sudden change in some established pattern of performance often indicates that a problem has developed. When employee turnover increases, sales decline, selling expenses increase, or more defective units are produced, a problem usually exists.
- Deviation from plan. When results do not meet planned objectives, a problem is likely. For example, a new product fails to meet its market share objective, profit levels are lower than planned, the production department is exceeding its budgets. These occurrences signal that some plan is off course.
- Out side criticism. The actions of outsiders may indicate problems. Customers may be dissatisfied with a new product or with their delivery schedules; a labor union may present a grievance; investment firms may not recommend the organization as a good investment opportunity; alumni may withdraw their support from an athletic program.

Decision makers face three types of problems:

- A crisis problem is a serious difficulty requiring immediate action. An example of a crisis is a severe cash flow deficiency that has a high potential of evolving into serious losses, and a customer protest against the quality of a product.
- A non-crisis problem is an issue that requires resolutions but does not simultaneously have the importance and immediacy characteristics of a crisis. Many of the decisions that managers make center on non-crisis problems. Example of such problems is a factory that needs to be brought into conformity with new state antipollution standards during the next three years and an employee who frequently is late for work.
- An opportunity problem is a situation that offers strong potential for significant organizational gain if appropriate actions are taken. Opportunities typically involve new ideas and novel directions that could be used, rather than difficulties that must be resolved. Non-innovative managers tend to focus on problems rather than opportunities.

Confusions are common in problem definition because the events or issues that attract the manager's attention may be symptoms of another more fundamental and pervasive difficulty than the problem itself. That is, there may exist confusion on the identification

of a problem and its symptoms. The accurate definition of a problem affects all the steps that follow. Managers once they have identified problems, they have to try to diagnose the cause of the problem. Causes unlike symptoms are seldom apparent.

This step has three general stages: scanning, categorization, and diagnosis.

- 1) **Scanning stage:** involves monitoring the work situation for changing circumstances that may signal the emergence of a problem. At this point the manager may be only vaguely aware that an environmental change could lead to a problem or that an existing situation constitutes a problem.
- 2) **Categorization stage:** entails attempting to understand and verify signs that there is some type of discrepancy between the current state and the desired state. At this point the manager attempts to categorize the situation as a problem and a no problem, even though it may be difficult to specify the exact nature of the problem, if one exists.
- 3) **Diagnosis stage:** involves gathering additional information and specifying both the nature and the causes of the problem. Without appropriate diagnosis, it is difficult to experience success in the rest of the decision-making process. At the diagnosis stage, the problem should be stated in terms of the discrepancy between current conditions and what is desired; the cause of the discrepancy should be specified.

2. Developing Alternatives

Before a decision is made feasible alternatives should be developed. This is a search process in which relevant internal and external environment of the organization are investigated to provide information that can be developed into possible alternatives. At this point it is necessary to list as many possible alternative solutions to the problem as you can. No major decision should be made until several alternative solutions have been developed. Decision-making at this stage requires finding creative and imaginative alternatives using full mental faculty. The manager needs help in this situation through brainstorming or Delphi technique.

3. Evaluating Alternatives

Once managers have developed a set of alternatives, they must evaluate them to see how effective each would be. Each alternative must be judged in light of the goals and resources of the organization and how well the alternative will help solve the problem. In addition, each alternative must be judged in terms of its consequences for the organization. Will any problems arise when a particular course of action is followed? Such factors as worker's willingness...

4. Choosing an Alternative

Based on the evaluation made managers select the best alternative. In trying to select an alternative or combination of alternatives, managers find a solution that appears to offer the fewest serious disadvantages and the most advantages. The purpose of selecting an alternative is to solve the problem so as to achieve a predetermined objective. Managers should take care not to solve one problem and create another with their choice.

A decision is not an end by itself but only a means to an end. This means the factors that lead to implementation and follow-up should follow solution selection.

5. Implementing and Monitoring the Chosen Solution

For the entire decision-making process to be successful, considerable thought must be given to implementing and monitoring the chosen solution. It is possible to make a "good" decision in terms of the first five steps and still have the process fail because of difficulties at this final step.

Implementing the Solution: A decision that is not implemented is little more than an abstraction. In other words, any decision must be effectively implemented to achieve the objectives for which it was made. Implementing a decision involves more than giving orders. Resources must be acquired and allocated. Decisions are not ends by themselves they are means to an end; so proper implementation is necessary to achieve that end.

Monitoring the solution: Monitoring is necessary to ensure that things are progressing as planned and that the problem that triggered the decision process has been resolved. Effective management involves periodic measurements of results. Actual results are compared with planned results (the objective); if deviations exist, changes must be made. Here again we see the importance of measurable objectives. If such objectives do not exist, then there is no way to judge performance. If actual results do not much planned results, then the changes must be made in the solution chosen, in its implementation, or in the original objective if it deemed unattainable. The various actions taken to implement a decision must be monitored. The more important the problem, the greater the effort that needs to be expended on appropriate follow up mechanisms. Are things working according to plan? What is happening in the internal and external environments as a result of the decision? Are subordinates performing according to expectations? must be closely monitored.

Decision-Making Conditions

When managers make decisions, the amount of information available to them or the degree of knowledge they have about the likelihood of the occurrence of each alternative vary from managers to managers or/and from situation to situation. To put it in other way, decisions are made under three basic conditions. These are condition of certainty, condition of risk, and condition of uncertainty.

1. Decision-making under Certainty

When managers know with certainty what their alternatives are and what conditions are associated with each alternative, a state of certainty exists. Decisions under certainty are those in which the external conditions are identified and very predictable; i.e. we are reasonably sure what will happen when we make a decision. The information is available and is considered to be reliable, and we know the cause and effect relationships. In decision-making under certainty there is a little ambiguity and relatively low chance of making poor/bad decisions. Decision-making under certainty seldom occurs, however, because external conditions seldom are perfectly predictable and because it is impossible to try to account for all possible influences on any given outcome it is very rare.

2. Decision-making under Risk

A more common decision-making situation is under risk. Under the state of risk, the availability of each alternative, the likelihood of its occurrence and its potential payoffs and costs are associated with probability estimates; i.e. decisions under risk are those in which probabilities can be assigned to the expected outcomes of each alternative. In a risk situation, managers may have factual information, but it may be incomplete. There is moderate ambiguity and moderate chance of making bad decision. E.g. tossing a coin, metrology

3. Decision-making under Uncertainty

Under this condition the decision maker does not know what all the alternatives are, what the probability of each will occur is or what consequences each is likely to have. This uncertainty comes from the dynamism of contemporary organizations and their environment. Big multi-national corporations assume these kinds of decisions. Decision-making under uncertainty is the most ambiguous and there is high chance of making poor decisions. In decision-making under uncertainty, probabilities cannot be assigned to surrounding conditions such as competition, government regulations, technological advances, the over all economy, etc. Uncertainty is associated with the consequences of alternatives, not the alternatives themselves. The decision-making is like being a pioneer. Reliance on experience, judgment, and other people's experiences can assist the manager is assessing the value of alternatives. E.g. Innovation of new machine, journey of discoverers.

Types of Decisions

Decisions can be classified in to: programmed and non programmed.

1. Programmed Decisions

Programmed decisions are those made in routine, repetitive, well-structured situations through the use of predetermined decision rules. The decision rules may be based on habit, computational techniques, or established policies and procedures. Such rules usually stem from prior experience or technical knowledge about what works in the particular type of situation. Most of the decisions made by first line managers and many of those made by middle managers are the programmed type, but very few of the decisions made by top-level managers are the programmed type. Managers can usually handle programmed decisions through rules, procedures, and policies.

E.g. Establishing a re-order point, Decide if students meet graduation requirements, Determination of employee pay rates

2. Non-programmed Decisions

Non-programmed decisions are used to solve non-recurring, novel, and unstructured problems. No well-established procedure exists for handling them, because it has not occurred before managers do not have experience to draw up on, or problems are complex or completely new. Because of their nature non-programmed decisions usually involve significant amounts of uncertainty. They are treated through farsightedness. Most of the highly significant decisions that managers make fall into the non-programmed

category. Non-programmed decisions are commonly found at the middle and top levels of management and are often related to an organization's policy-making activities.

E.g. To add a product to the existing product line, to reorganize a company, to acquire another firm

Types of Managerial Decisions

Type of decisions	Type of problem	Procedures	Examples
Programmed	Repetitive, routine	Rules, standard operating procedures, policies	Business: processing payroll vouchers College: processing admission applicants Hospital: preparing patient for surgery. Government: using state owned motor vehicle.
Non-programmed	Complex, novel	Creative problem solving	Business: introducing a new product. College: constructing new classroom facilities Hospital: reacting to regional disease epidemic Government: solving spiraling inflation problem

In reality most decisions fall between the two; i.e. a continuum of decision situations exists ranging from those that are highly structured to those that are unstructured. Situations between the two extremes are partially structured. As the name suggests, in a partially structured situation, only a part is well structured. Typically, although the manager has a great deal of data available, the final choice is not obvious. Many intangibles are involved in the final choice. Therefore, the manager must base the ultimate decision on the data and supplementary factors, using judgment and experience.

E.g. A hospital wishing to improve patient care may adjust its patient-staff ratio (programmable situation), reorganize its staff (a non programmable situation).

Continuum of Decision situations

WELL STRUCTURED (PROGRAMMED)

1. Specification of decision procedure agreed in advance of resolution.
2. Little managerial involvement at time of each resolution
3. Repeated resolutions with same data yield same results.

PARTIALLY STRUCTURED

1. Only a part of decision process can be completely specified and structured.
2. Manager makes final resolution from structured portion of his/her experience and from intuition.
3. Different managers may agree on certain data

ILL-STRUCTURED (NON PROGRAMMED)

1. Decision procedure can not be completely structured in advance of resolution.
2. Individuals resolve each situation on the basis of experience and judgment.
3. Different managers may reach different conclusions.

Why Do Managers Make Poor Decisions?

All managers recognize the importance of making sound decisions. Yet most managers readily admit having made poor decisions that hurt their company or their own effectiveness. Why do managers make mistakes? Why don't decision always result in achieving some desired goal? Making the wrong decision can result from any one of these decision-making errors:

- **Lack of adequate time**
Waiting until the last minute to make a decision often prevents considering all alternatives. It also hampers thorough analyses of the alternatives.
- **Failure to define goals**
Objectives cannot be attained unless they are clearly defined. They should be explicitly stated so that the manager can see the relationship between a decision and a desired result.
- **Using unreliable sources of information**
A decision is only as good as the information on which it is based. Poor sources of information always result in poor decisions.

- **Fear of consequences**

Managers often are reluctant to make bold, comprehensive decisions because they fear disastrous results. A “play it safe” attitude sometimes limits a manager’s effectiveness.

- **Focusing on symptoms rather than causes**

Addressing the symptoms of a problem will not solve it. Taking aspirin for a toothache may provide temporary relief, but if an abscess causes the pain, the problem will persist. Business managers too often foul on the results of problems instead of the causes.

- **Reliance on Hunch and Intuition**

Intuition, judgment and ‘feel’ are important assets to the decision maker. But a manager who permits intuition to outweigh scientific evidence is likely to make a poor decision.

Some times a manager’s decision is not exactly “poor”, but it still doesn’t produce optimal results. Less than optimal decisions can have three causes:

1. **Bounded rationality** imposes limits on a decision, such as that it should be economical or logistically practical. This limit serves as a screening device, eliminating some of the alternatives. The manager must choose from the options that have filtered through the restrictions. The overall optimal decision may no longer be a valid option when using this method. The decision maker simply selects the best alternative, given various specifications that must be met.
2. **Sub optimization** is a manager’s tendency to operate solely in the interests of his/her department rather than in the interests of the company as a whole. In making a decision, the department manager cannot be so self-centered as to ignore the effects of the action on other areas. The key is to improve the company’s performance, not just the performance of one department.
3. **Unforeseen changes in the business environment** also cause less than optimal decisions.